

Thinking Outside the Simplex

Signed compositions of optimization algorithms:
six semantics, two traps, and two survivors

Peter Cotton

July 14, 2026

Abstract

A companion note [1] blends optimizers by placing them at the vertices of a simplex and letting a language model realize each recipe as a program. Here the weights may go negative, and the question is what a negative weight should mean. Six semantics are tested under a paired protocol: every generated program gates its negative-weight machinery through one constant, so the same program with the machinery disabled is scored on identical instances. Avoidance semantics wins selection and is reversed by its ablation. Caricature semantics, exaggerate the host along the axes where it most differs from the negated reference, is the only one to beat its twin at selection, and a search of the extended space finds a program at regret 0.047 against its twin’s 0.181. On problems never used in selection that comparison reverses, the twin winning 65% of paired instances ($p = 10^{-11}$); yet on weaker hosts the machinery survives the same test (up to 65% for the star, $p = 10^{-12}$). The pattern: negative weights repair weak hosts and spoil strong ones. The byproduct stands either way: the twins of the strongest recipes twice rank first at the largest budget among fourteen optimizers.

1 The construction, and six meanings of a minus sign

A simplex recipe over K algorithms has weights $w_k \geq 0$ summing to 1. Drop the non-negativity and keep the sum: recipes now range over an affine slice of which the simplex is the all-positive face.

A negative weight needs a semantics, and six were tested. Avoidance: shadow the negated algorithm’s proposal logic at zero evaluation cost and steer away from where it would go. Density: thin the host’s proposals toward the signed mixture of proposal densities, by rejection, before evaluating. Caricature: exaggerate the host along the axes where it most differs from the negated reference, extrapolation in design space. Pitfalls: design the negated algorithm’s characteristic failure modes out of the host. Baseline: treat the negated algorithm’s shadow as the bar to beat, escalating when behind. Residual: strip from the negated algorithm everything the host already has and graft in what remains, amplified.

Attribution is by paired ablation. Every generated program must gate all its negative-weight machinery through one module constant, so its twin, the same program with the constant zeroed, recovers the plain host and is scored on identical instances. The paired delta, signed minus twin, is the only statistic that counts; single-draw leaderboards reward whatever a label elicits from the generator, not what the label claims.

2 Avoidance, and the first trap

Under avoidance semantics, the recipe +200% Nelder–Mead, −100% Differential Evolution won a 22-recipe selection round at regret 0.179 against 0.326 for the best unsigned control, and the generated program faithfully implements the semantics: a shadow DE population evolved at zero cost, a repulsion radius, penalty and push-away logic.

The ablation reverses it. Disabling the repulsion in that same program improves it to 0.130 on identical instances. The win belonged to the unusually good Nelder–Mead host the prompt elicited; the repulsion was a handicap, out of sample as well, where the ablated twin also ranks better at every budget.

3 The bake-off

Table 1 gives the mean paired delta per semantics on the pairs predicted favourable, from two draws per recipe on a selection suite spanning 2 to 100 dimensions. Caricature is the only semantics that helps on average, and the only one with an artifact that decisively beats its twin (0.104 against 0.181). The mathematically prettiest semantics, rejection-thinned proposal densities, goes zero for five against its twins.

semantics	mean paired delta	arms improving
caricature	−0.032	4/8
repulsion	−0.009	3/7
residual	+0.014	2/8
pitfalls	+0.018	2/7
density	+0.040	0/5
baseline	+0.086	0/4

Table 1: Six semantics under the paired protocol; negative delta means the signed machinery helps on the selection suite.

Out of sample the bake-off winner is even with its twin (53% paired, $p = 0.17$): the selection-suite advantage neither transfers nor reverses. Both versions rank first and second of fourteen at the largest budget on the held-out problems, ahead of the champion blend of the companion note.

4 The map and the dial

Sweeping all twenty ordered pairs under caricature at unit magnitude, eight of nineteen scored pairs benefit; the largest paired gains are simulated annealing exaggerated away from pattern search (−0.24) and CMA-ES away from Differential Evolution (−0.17).

The ancestor rank-correlation matrix does not predict the map: across the nineteen pairs, paired delta and ancestor correlation are uncorrelated (+0.05). Correlation harvesting is the mathematics of runtime subtraction; caricature is a generation-time act, and the map must be measured, not derived.

The magnitude dial has an optimum at restraint. On the strongest pair, a mild caricature ($\lambda = 0.5$) gives the best paired deltas (−0.02, −0.13); at $\lambda = 2$ and beyond the exaggerations

```

Build a black-box numerical optimizer from the SIGNED recipe: NelderMead
+59%, DifferentialEvolution -108%, CMAEvolutionStrategy -13%,
PatternSearch +169%, SimulatedAnnealing -8%

HOST (74% of positive mass): PatternSearch - Hooke-Jeeves coordinate moves
with a step size that shrinks on failure. Build the architecture from this.
GRAFT into it: 25% NelderMead (downhill simplex: reflect/expand/contract/
shrink a set of n+1 points)

Negative weights mean CARICATURE, extrapolation in design space: for each
reference below, identify the axes on which the host most differs from it
and EXAGGERATE the host along exactly those axes, with the size of each
exaggeration proportional to the stated lam times ANTI_STRENGTH.
CARICATURE AWAY FROM DifferentialEvolution (lam = 1.08): population +
difference-vector mutation (rand/1, current-to-best/1) and crossover
CARICATURE AWAY FROM CMAEvolutionStrategy (lam = 0.13): sample from a
Gaussian whose covariance adapts to successful steps
CARICATURE AWAY FROM SimulatedAnnealing (lam = 0.08): accept uphill
moves with probability  $\exp(-dF/T)$ ; cool T over time

[followed by the mandatory ANTI_STRENGTH gate and the interface contract]

```

Figure 1: The prompt at the best point found in the extended space.

tip into parody and the deltas turn positive.

5 The extended space, searched, and the second trap

Sampling the full signed slice (weights in roughly $[-1, 2]$, positive part blending hosts, each negative weight a caricature reference with its magnitude as exaggeration length) produced the best selection-stage artifact of the entire research program: regret 0.047, twin 0.181, paired delta -0.134 , at the recipe of Figure 1. On the selection suite the caricature machinery is unambiguously causal.

On twenty-nine problems never used in any selection step, the comparison reverses. The twin beats the signed original on 65% of paired instances ($p = 9 \times 10^{-12}$), ranks first of fourteen at the largest budget (4.5 against the signed version’s 6.5), and the signed version also loses to the companion note’s champion (41%, $p = 3 \times 10^{-5}$).

This is the second trap, and it is not the first one again. The paired ablation solved attribution: the caricature machinery really did cause the selection-suite gain. What it did not solve is generalization: the credited machinery was itself tuned, by selection, to the suite. Attribution and transfer are different walls, and a delta earned on the selection distribution is a claim about that distribution only.

6 What survives

Racing the sweep’s other winners the same way completes the picture, and it is not a uniform failure. The CMA-ES-away-from-DE artifact beats its twin on 65% of held-out paired instances ($p = 2 \times 10^{-12}$), and the annealing-away-from-pattern-search artifact on 59% ($p = 10^{-5}$): the caricature machinery survives out of sample on these hosts. The protocol still acquires its second stage, since two other winners did not survive it: paired ablations must be scored out of sample before a delta counts.

The pattern across the four raced winners is consistent. Where the twin is already excellent, pattern-search hosts ranking first at the largest budget, the signed term vanishes or reverses; where the twin is weak, CMA and annealing hosts in the bottom half of the field, the signed term repairs it and survives. Negative weights, under the one semantics that works, are a treatment for weak hosts, not an enhancement for strong ones, and no signed artifact yet leads the held-out field absolutely.

The byproduct stands either way, twice replicated: signed prompts elicit excellent conventional programs. The twins of the two strongest recipes, both budget-aware pattern-search hosts with Nelder–Mead grafts, rank first of fourteen at the largest budget on the held-out problems, ahead of every prior artifact. Exploring outside the simplex keeps producing better inhabitants of the inside.

Reproducibility. Harnesses `e13_signed.py`, `e15_semantics.py`, `e16_caricature_sweep.py` and `e17_search.py` under `papers/dfo_recommender/` in the humpday repository [2]; runs and every generated program are in the adjacent `runs/` directories.

References

- [1] P. Cotton. Alloy: a machine-designed derivative-free optimizer, and a full account of the search that found it. Working paper, <https://humpday.microprediction.org/papers.html>, 2026.
- [2] P. Cotton. HumpDay: recommendations and comparisons of derivative-free optimizers. <https://humpday.microprediction.org>, 2026.